

Model	Random	Link	systematic
Linear Regression	normal	Identity	continuous
Logistic Regression	Binomial	Logit	Mixed
Poisson Regression	Poisson	Log	Mixed

Random: probability distribution of Response (Y)

Link function:

Identity: $g(\lambda(x)) = \lambda = \beta_0 + \beta_1 x = E(Y_i), Y_i \sim N(\lambda_i, \sigma^2)$

logit: $\text{logit}(\lambda(x)) = g(\lambda(x)) = \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x$

π : probability of success, $Y_i \sim \text{Bin}(n_i, \pi_i)$

log: $g(\lambda(x)) = \log(\lambda) = \beta_0 + \beta_1 x, Y_i \sim \text{Poisson}(\lambda_i)$

$$\lambda = e^{\beta_0 + \beta_1 x}$$

logit model

$$Pr(Y=1|x) = \frac{1}{1+e^{-\beta x}}$$

" CDF of logit function

probit model

$$Pr(Y=1|x) = \Phi(x^T \beta) \quad (\text{CDF of normal func})$$

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

cost $J_{\text{ML}} = \sum_i y_i \log\left(\frac{1}{1+e^{-\beta x_i}}\right) + (1-y_i) \log\left(1 - \frac{1}{1+e^{-\beta x_i}}\right)$

$J_{\text{ML}} = \sum_i y_i \log \Phi(x_i^T \beta) + (1-y_i) \log (1 - \Phi(x_i^T \beta))$

Logistic / Probit Regression

* generalized linear model (p features)

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_p x_p$$

* logit link fun

$$h_0(x) = \ln\left(\frac{p}{1-p}\right) \quad \text{odds}$$

= logit(p)

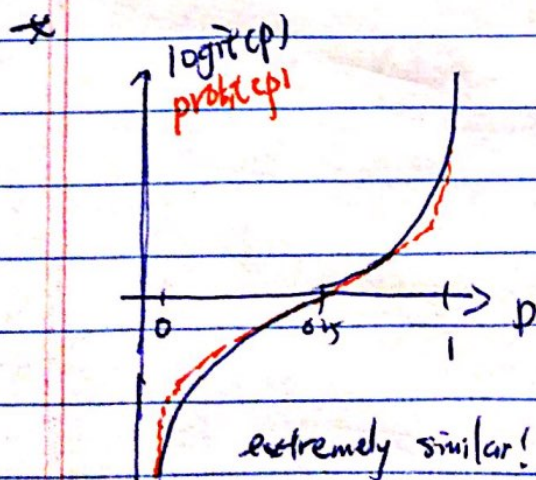
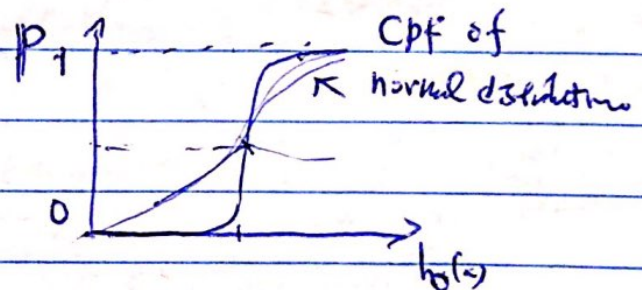
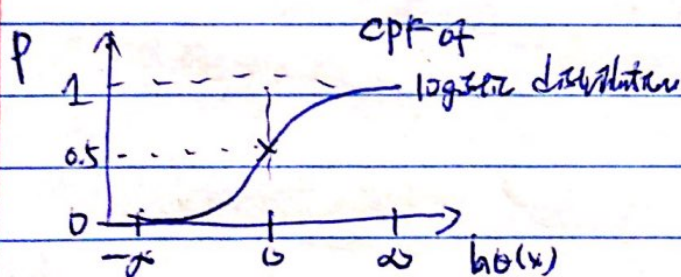
$$p = \text{logit}^{-1}(h_0(x)) = \frac{1}{1 + e^{-h_0(x)}} = P(Y=1|x)$$

inverse normal link fun

$$h_0(x) = \Phi^{-1}(p) \Rightarrow \Phi(h_0(x)) = P(Y=1|x)$$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{y^2}{2}} dy \quad (\text{CDF of normal})$$

So $h_0(x) = \theta_0 + \theta_1 x_1 + \dots + \theta_p x_p$



for any cumulative probability, what is the corresponding Z value?

